A Molecular Dynamics Study in Structural Dynamics of a V717I Substitution in the Amyloid Precursor Protein

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1. Introduction

The effect of Valine to Isoleucine substitution at codon 717 in Amyloid Precursor Protein (APP) [1] is of great importance to understand the stability and properties of mutant beta amyloid peptides in Alzheimer's disease. Alzheimer's disease is a neurodegenerative disorder which leads to the death of nerve cells and hence gradually reduces memory and cognition. One of the prominent features of Alzheimer's disease is the build-up of plaques between nerve cells which are formed when beta amyloid cluster together. These beta amyloids are formed when enzymes act on the APP and are involved in the ability of neurons to change and adapt over time. We have performed Molecular Dynamics (MD) simulations of the wildtype and mutant transmembrane domain of amyloid precursor protein interacting with a model POPC membrane. Here, we present the study in structural dynamics of both mutant and wildtype APP.

2.Wildtype

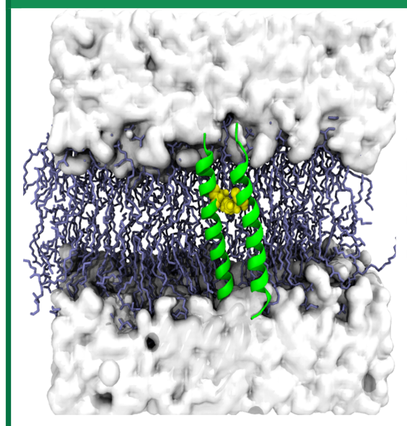


Fig.1. Wild type protein inside POPC bilayer

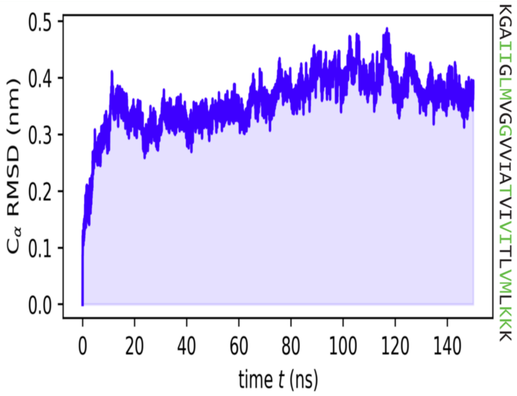


Fig. 2. Root Mean Square Deviation (RMSD) of C-alpha atoms

Fig.3. Protein contact map for wildtype

- In Fig.1. Wildtype protein (green), represented by cartoon representation is inside the POPC bilayer (light blue), represented by licorice representation with water (white) in Quick Surf representation and Valine residue in VDW representation.
- In Fig.2. RMSD plot for wildtype shows that the RMSD converges after 100 ns.
- In Fig.3. Protein contact map is shown for two peptides in dimer where the darker square shows the closer the two amino acids are. Amino acids in green are the ones interacting while the ones in black are not interacting.

3. Mutant

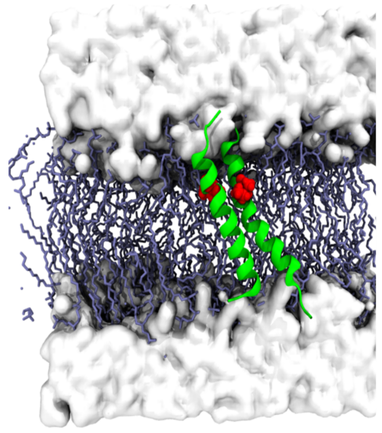


Fig.4. Mutant Protein inside POPC bilayer

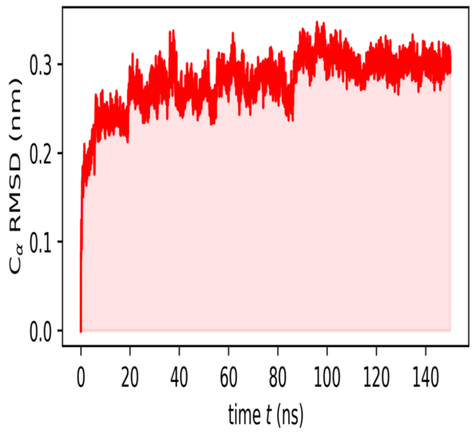
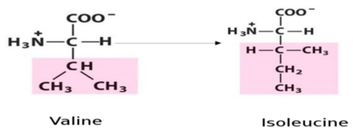


Fig.5. Root Mean Square Deviation (RMSD) of C-alpha atoms

Fig.6. Protein contact map for mutant



- In Fig.1. Mutant protein (green), represented by cartoon representation is inside the POPC bilayer (light blue), represented by licorice representation with water (white) in Quick Surf representation and Isoleucine residue in VDW representation.
- In Fig.2. RMSD plot for mutant shows that the RMSD converges after 100 ns.
- In Fig.3. Protein contact map is shown for two peptides in dimer where the darker square shows the closer the two amino acids are. Amino acids in green are the ones interacting while the ones in black are not interacting.

4. Analysis

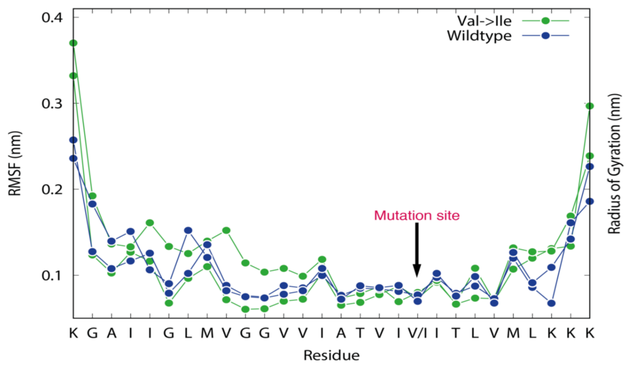


Fig. 7. Root Mean Square Fluctuation

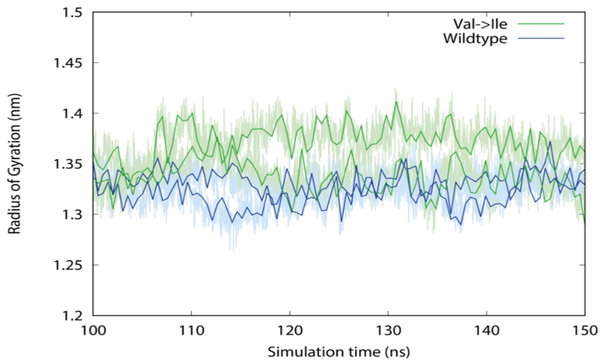


Fig. 8. Radius of Gyration

- On substituting the valine for an isoleucine, there was a noticeable increase in helix-helix packing, as per the distances measured in an amino-acid contact map.
- The radius of gyration has a slight increase in at least one of the two mutant peptides as isoleucine has a longer/bigger side chain than valine.
- There is not much change in fluctuation of valine and isoleucine at the mutation site.
- Given the amino acid sequence of a transmembrane (TM) helix, the server [2] gives a prediction for free energy of TM helix insertion which for our wildtype protein is estimated as -1.30 Kcal/mol and for mutant as -1.59 Kcal/mol.

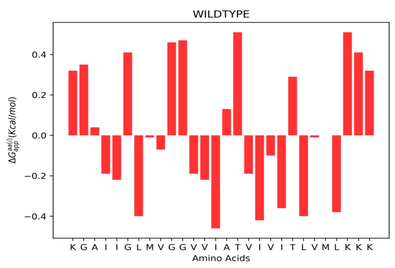


Fig. 9. ΔG contribution from each amino acid in wildtype

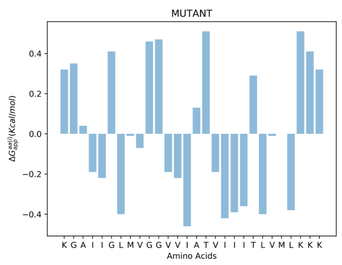


Fig. 10. ΔG contribution from each amino acid in mutant

5. Conclusions

We performed the Molecular Dynamics simulation of APP inside POPC bilayer for both wildtype and mutant and all the analysis works were peformed over 50 ns from 100 ns - 150 ns.

- Simulations were shown to converge after 100 ns of computational time.
- The position of the substitution was at a different orientation relative to the helix-helix interface, positioned within the bilayer lipid chains.
- The substitution resulted in a variation in helix-helix side-chain contacts.
- Our preliminary data would suggest a change in APP structural dynamics, which may impact on how the dimer forms, the rate of its association and how readily it is to leave the bilayer after being enzymatically cut.
- We are planning to construct a free energy profile using metadynamics.

6. References

[1] Chen, W., Gamache, E., Rosenman, D. J., Xie, J., Lopez, M. M., Li, Y. M., Wang, C. **(2014)**. Familial Alzheimer's mutations within APPTM increase A42 production by enhancing accessibility of -cleavage site. Nature communications, **5**, 3037.

[2] Hessa, T., Meindl-Beinker, N., Bernsel, A., Kim, J., Sato, Y., Lerch, M., Lundin, C., Nilsson, I., White, SH. and von Heijne, G. **(2007)** Molecular code for transmembrane-helix recognition by the Sec61 translocon. Nature. **450**, 1026-1030.